



10CFR 50.73

CCN: 18-60

June 21, 2018

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Peach Bottom Atomic Power Station (PBAPS) Unit 3
Renewed Facility Operating License No. DPR-56
NRC Docket No. 50-278

Subject: Licensee Event Report (LER) 3-18-001

Enclosed is a Licensee Event Report concerning a condition prohibited by Technical Specifications. In accordance with NEI 99-04, the regulatory commitment contained in this correspondence is to restore compliance with the regulations. The specific methods that have been planned to restore and maintain compliance are discussed in the LER. If you have any questions or require additional information, please do not hesitate to contact Jim Kovalchick at 717-456-3351.

Sincerely,

A handwritten signature in black ink, appearing to read "P. Navin", written over a light gray background.

Patrick D. Navin
Site Vice President
Peach Bottom Atomic Power Station

PDN/dnd/IR 4129583

Enclosure

cc: US NRC, Administrator, Region I
US NRC, Senior Resident Inspector
R. R. Janati, Commonwealth of Pennsylvania
S. Gray, State of Maryland
B. Watkins, PSE&G, Financial Controls and Co-Owner Affairs



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Peach Bottom Atomic Power Station Unit 3

2. Docket Number

05000278

3. Page

1 OF 3

4. Title

Reactor Core Isolation Cooling System Pressure Switch Failure Results in Condition Prohibited by TS

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
04	22	2018	2018	- 001	- 0	06	21	2018	Facility Name	Docket Number 05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100%	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact

James M. Kovalchick, Regulatory Assurance Manager

Telephone Number (Include Area Code)

717-456-3351

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
D	BN	PS	S382	Yes					

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On 4/22/18, during a routine surveillance test of the Reactor Core Isolation Cooling (RCIC) system, the RCIC pump turbine tripped shortly after startup. The trip was the result of an indication of high pressure in the RCIC turbine discharge line. Troubleshooting determined that a pressure switch that monitors turbine exhaust pressure had failed. Actual exhaust pressure during the test was at the expected pressure of approximately 12 psig.

The pressure switch was replaced and the RCIC system was restored to an operable status on 4/23/18. There were no actual safety consequences as a result of this condition.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Peach Bottom Atomic Power Station Unit 3	05000278	2018	- 001	- 0

NARRATIVE**Unit Conditions Prior to the Event**

Unit 3 was operating in Mode 1 at approximately 100% rated thermal power. There were no structures, systems or components out of service that contributed to this event.

Description of Event

On 4/22/18, during a routine surveillance test of the Reactor Core Isolation Cooling (RCIC) system [EIS:BN], the RCIC pump turbine [EIS:TRB] tripped approximately 28 seconds after startup, prior to the system reaching rated flow and pressure. Concurrent with the RCIC trip, an alarm was received for RCIC turbine high exhaust pressure. Local indication for the exhaust line pressure was at the expected pressure of approximately 12 psig, below the trip setpoint of 50 psig, which is active after a 15-second time delay. Operators verified proper system response following the RCIC turbine trip.

The RCIC system was declared inoperable at approximately 9:12 am on 4/22/18. Technical Specification (TS) 3.5.3 Condition A was entered, which requires the RCIC system to be restored to operable within 14 days. Initial troubleshooting determined a RCIC turbine exhaust pressure switch [EIS:PS] had failed. At approximately 8:55 pm on 4/22/18, the pressure switch was electrically isolated, which made the RCIC system available to perform its function. After the switch was replaced, a surveillance test was successfully completed at 2:53 pm on 4/23/18, allowing the RCIC system to be returned to an operable status. The RCIC system was inoperable for approximately 29 hours and 19 minutes.

Analysis of Event

The RCIC system provides adequate core cooling and control of the Reactor Pressure Vessel (RPV) water level following an RPV isolation accompanied by a loss of coolant flow from the feedwater system. The High Pressure Coolant Injection (HPCI) system can also perform this function and was available during this event.

The trip of the RCIC turbine on 4/22/18 was the result of an indication of high pressure in the RCIC turbine exhaust line. Two pressure switches monitor this pressure and are designed to shut down the turbine if discharge pressure reaches 50 psig. Actual exhaust pressure during the test was at the expected pressure of approximately 12 psig. Troubleshooting determined one of the two pressure switches had failed, which resulted in the trip of the RCIC pump turbine.

The previous RCIC surveillance test was successfully completed on 1/16/18 which demonstrated operability. A degradation process, described in the Cause of the Event section of this report, had been occurring for a period of time. At some time between the successful test on 1/16/18 and the time at which the trip occurred during the surveillance test on 4/22/18, the degradation advanced to the point where an electrical short existed in the pressure switch. Although it is not known when this occurred, it is possible that

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CONTINUATION SHEET**

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1. FACILITY NAME

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2. DOCKET NUMBER

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2018	- 001	- 0

NARRATIVE

it occurred before 4/8/18, which would have resulted in the RCIC system being inoperable for a period of time exceeding the TS 3.5.3 allowed completion time of 14 days.

As a result, this event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B), as a condition prohibited by Technical Specifications. If the RCIC system is inoperable, TS 3.5.3 Condition A requires the system to be restored to an operable status within 14 days. There is not firm evidence for when the failure occurred, however, for the purposes of conservative reporting per 10 CFR 50.73, it is considered possible to have been greater than 14 days.

Cause of the Event

Further investigation determined the switch failed due to an electrical short within the switch. Water had entered the switch enclosure, which resulted in corrosion and degradation of the switch internals, causing the electrical short. A diaphragm normally isolates the switch from the instrument line that contains condensed steam from the RCIC turbine exhaust piping. A tear in the diaphragm resulted in a small amount of water entering the switch enclosure.

Discoloration of the interior of the metal switch enclosure indicated the water had been in the switch for a period of time. The moisture caused corrosion and degradation of the switch, which resulted in a short circuit, causing the switch to fail.

Maintenance practices for the switch were reviewed and were determined to be consistent with industry standards for this type of switch. In addition to the quarterly surveillance test, calibration of the switch is verified every 18 months. Calibration history was reviewed and the last time the switch was out of tolerance was in 2001. The switch was within normal tolerances with no adjustment necessary since that time. There were no adverse trends in the data reviewed.

Corrective Actions

The switch has been replaced. Maintenance performed for this type of switch is being evaluated to identify improvements that will identify this type of degradation. Additional corrective actions are documented in the corrective action program.

Previous Similar Occurrences

No previous similar occurrences have been identified.